Make Grove Talk Without a Large LLM

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The system can hold a useful English conversation on this laptop without a large LLM. To do it, it has to copy what every successful pre-LLM chatbot did: lead the conversation inside a narrow persona, choose replies from a curated bank instead of generating them, and hand every number to an exact engine. Those parts are a rule-and-embedding router, ChatScript-style topics with a no-repeat memory, a few thousand validated replies, and templates. On published desktop benchmarks they cost milliseconds per turn and well under 100 MB, and none needs a model larger than about 30M parameters. The hard limit is elsewhere. **Understanding word problems needs a learned program writer.** cpulm's 18.0% on SVAMP is level with a published bag-of-words baseline (18.3%). Tree solvers trained on similar data reach only 41–44%. The strongest small-scale evidence is a 0.77B code model that writes programs (51.0% on SVAMP) and a 1.3B generator plus a 1.3B verifier (81.5% on GSM8K). Grove changes the economics, because the system can execute several sampled programs and answer only when they agree, which turns a weak writer into a cautious one. **Open-domain chat and general questions are out of reach for cheap components.** Developer-reported scores drop sharply between 0.6B and 1.7B parameters, so these requests should be scoped out, deflected honestly, or passed to a local 1.7B model or to Neo. The six-stage plan at the end starts by measuring the Core Ultra 7 255U. **No benchmark cited in this report was run on that chip**, so every laptop latency below is an extrapolation from other machines.

> **How to read the numbers. [measured]** means an experimental result in a cited study, or the team's own run, on hardware other than the 255U unless stated. **[vendor]** means a figure reported by the developer of a model, library, product or chip about its own work: model cards, developer blogs, tech-report tables and marketing. **[community]** means hobbyist or automated benchmark sites. **[estimate]** means our extrapolation, which nothing has measured. Team results without a link are the team's own runs, not published elsewhere.

Pre-LLM chatbots succeeded by steering the conversation, not by understanding it

The rule-based lineage runs from ELIZA to Mitsuku, and its mechanics have barely changed. ELIZA ranked keywords, picked a decomposition rule for the highest-ranked keyword, reassembled fragments of the input with the pronouns swapped, and fell back to a content-free remark when nothing matched (Wikipedia: ELIZA). ChatScript, the engine behind several Loebner Prize winners, organised this machinery into **topics** that become candidates through keywords. Each topic holds three kinds of rule: responders that match user input, **gambits** the bot says when it has the floor, and **rejoinders** that match only the user's answer to the bot's last line. When nothing fires, the engine falls back to a quibble or a gambit. Patterns match lemmas and concept sets. A rule is erased once it fires, and any output used in the last 20 volleys is suppressed, which is how ChatScript avoided the repetition typical of AIML bots (Wilcox, "Winning the Loebner's"). Wilcox says the engine computes a reply in a few milliseconds **[vendor]**. He also says plainly where the cost lies. Prize winners carried about 11K–16K rules, ALICE about 130K and Mitsuku about 220K, and a typical entrant had spent more than five years building the bot's character. On Wilcox's own estimate, 10K–16K rules sustained roughly 10–32 hours of conversation before gambits ran out (same source). He also identifies the basic brittleness: a general pattern misfires and a specific pattern misses, so quality tracks the number of rules authored.

Task-oriented systems solved a different problem with the same frugality. Since GUS in 1977, frame systems have filled slots and asked about whichever slot is missing (Stanford CS224S slides). Rasa split this into NLU, a per-conversation tracker holding slots and an event log, and a policy that sees only intents and entities, never raw text (Bocklisch et al. 2017). Narrow-domain NLU is essentially solved. DIET reports that ATIS and SNIPS are close to 100%, reaches **90.18 intent F1** on the 64-intent NLU-Benchmark, and needs about 80 ms per

utterance, though the paper does not say whether that inference ran on its CPUs or its K80 GPU (Bunk et al. 2020) **[measured]**. Multi-domain state tracking never got there: TRADE reached only **48.62% joint goal accuracy** on MultiWOZ (ACL P19-1078). One task frame at a time is robust. Juggling several is not. Secondary sources describe Rasa Open Source as now in maintenance mode, so the team should copy the pattern, not adopt the framework.

Retrieval over a curated response set was the most successful commercial technique of the period. Google's Smart Reply never generated free text. It scored a fixed, human-validated response set and handled **10% of mobile replies** in Inbox (Kannan et al. 2016). About 70% of XiaoIce's replies were retrieved from its own past conversations. In Microsoft's human ratings of 5,000 dialogues, the retrieval-only system was rated unacceptable 35.0% of the time, generation alone 66.6%, and the hybrid that ranked both together 23.1% (Zhou et al. 2020) **[vendor]**. Retrieval gave high quality with low coverage, and generation patched the coverage. The ranker can be cheap. On Ubuntu next-utterance selection, plain TF-IDF picks the right reply from 10 candidates **41.0%** of the time, the same as a vanilla RNN, against 60.4% for an LSTM dual encoder (Lowe et al. 2015) **[measured]**.

The Alexa Prize socialbots show both how far hand engineering goes and where it stops. The 2018 winner, Gunrock, routed input to 11 topic modules. It filled templates from knowledge-base queries, rotated several templates per state without repeats, and paired each acknowledgement with a follow-up question. After the competition it averaged a 3.65 user rating over 34,432 conversations. Users who asked about the bot's backstory gave notably higher ratings (Yu et al. 2019) **[measured]**. Winning ratings rose only from 3.17 in 2017 to 3.81 in 2020 (Voicebot.ai; Emory NLP), and no team ever met the Grand Challenge bar of a 4.0 rating and 20-minute conversations (Voicebot.ai). Together these results define what to build. Grove and the program translator are the "skills". The missing pieces are a dialogue manager that triggers skills and walks a fallback chain, a persona with an answerable backstory, a curated reply bank with retrieval, and a no-repeat memory. Without a generator, expect a ceiling of roughly 3–4 out of 5 on open small talk, and expect quality to grow with authored content rather than with model size.

Understanding is nearly free until a sentence hides a calculation

Intent routing and similarity cost almost nothing on a CPU. Static embeddings are lookup tables distilled from a transformer. Model2Vec's potion-base-8M encodes about **124,000 sentences per second** on an i7-13700K desktop CPU, against 1,739 for all-MiniLM-L6-v2 (HF blog: static embeddings) **[vendor]**. It gives up little where this system needs it. On MTEB classification it scores 70.34 against MiniLM's 69.25, and the loss is concentrated in retrieval (31.1 against 42.9) (Model2Vec results) **[vendor]**. With about ten examples per intent, frozen sentence encoders plus a light classifier reach roughly 83–93% on the 77-, 150- and 64-way BANKING77, CLINC150 and HWU64 benchmarks, and they train "in a matter of minutes on a single CPU" (Casanueva et al. 2020) **[measured]**; the per-benchmark cells came from search snippets and should be re-checked. A maths chat needs a few dozen intents, not 150, so a potion-base-8M embedding feeding scikit-learn logistic regression, with an out-of-scope threshold, is the right router. On the 255U it should cost well under 1 ms per message, with about 5–20 ms for MiniLM where paraphrase precision matters **[estimate]**.

Structural parsing is just as affordable. spaCy's CNN pipeline processes about **10,000 words per second** on CPU with 85.5 NER F1, against 684 words per second for its transformer pipeline and 878 for Stanza (spaCy facts & figures) **[vendor, hardware unstated]**. The small English model gives subject, verb and object links ("who gave what to whom") in a few milliseconds a sentence **[estimate]**. Number language such as "three dozen", "half as many" and "twice" is better handled by a deterministic normaliser feeding Grove than by NER. Tools such as text2num, quantulum3 and Pint exist, but the notes found no benchmarks for them.

The autocorrect needs one adjustment in this light. At about **3 ms per word [measured, team]**, a 15-word message spends about 45 ms in spelling correction, which makes it the slowest component in a pipeline whose other stages cost single milliseconds. Gating it to out-of-vocabulary tokens should cut most of that **[estimate]**. Its error profile also shapes the router. It fixes 61% of typos, so about four typos in ten reach the

classifier, which therefore needs subword or character features and typo-invariance tests. Its 0.04% false-change rate implies about 0.6% of 15-word messages get a correct word altered (15 × 0.0004) **[estimate]**. For names of people and objects, Gunrock's trick of matching Double Metaphone phonetic codes against a known-entity list complements a frequency-based corrector (Yu et al. 2019).

Word problems are where cheap understanding breaks down, and SVAMP explains why. Patel et al. built SVAMP from 1,000 variations of simple one-unknown problems. They showed that tree solvers scoring 83–89% on MAWPS still score **77.7% with the question deleted**, so they had learned surface cues, not the problem. Trained on MAWPS plus ASDiv-A, the best of them reached only **43.8% on SVAMP** (Graph2Tree with RoBERTa). A bag-of-words model with no word order scored **18.3%**, and the majority baseline 11.7%. Accuracy fell from 78.3% with two numbers in the problem to 25.4% with three, and inverting the operation was the hardest variation (Patel et al. 2021) **[measured]**. cpulm's **18.0% [measured, team]** sits on that bag-of-words line. The training data differ, so the match is not exact, but it is strong evidence that more pattern capacity will not help. The model needs sensitivity to the question sentence and to word order. Classic structured systems remain relevant. Roy and Roth's expression-tree parser combines quantity-relevance and operation classifiers with a constrained beam search, where answers must be positive and integral for counts. It scored 78.0% on single-step add/subtract problems and 45.2% on multi-step ones, against 2.3% for template matching (Roy & Roth 2016) **[measured]**.

The strongest small-scale evidence points to pretrained code models trained to emit programs. Program-aided Distillation fine-tuned CodeT5-large (0.77B) on execution-filtered programs and reached **51.0% on SVAMP and 44.9% on GSM8K**, against 7.2% and 7.5% with standard fine-tuning. Even CodeT5-small, at **60M parameters**, reached 32.2% on GSM8K. With the same teacher and data, program distillation beat chain-of-thought distillation on SVAMP by 36.7% to 18.6% (PaD) **[measured]**. Qwen2.5-0.5B-Instruct reports 49.6% on GSM8K and the 1.5B model 73.2% (Qwen2.5 blog) **[vendor]**. cpulm is 10.7M parameters trained from scratch. The closest working recipe starts from a model about 6x larger that already knows English and code, and trains it on verified programs. The gap is pretraining, not architecture.

Fluent replies come from templates or from about 1.7B parameters, with little in between

Templates are the safe floor, and the best controlled comparison supports them more than their reputation suggests. In the E2E NLG Challenge, 21 systems realised restaurant meaning representations. Hand-engineered systems lost to neural seq2seq on word-overlap metrics and on rated naturalness, but matched or beat them on overall human-rated quality and on output complexity and diversity. Vanilla seq2seq systems often failed to express the given meaning unless decoding had a semantic control step such as a reranker or slot check (Dušek, Novikova & Rieser 2020) **[measured]**. Grove plays the role of the meaning representation, and a template realiser guarantees its numbers come out exactly. Templates sound robotic mainly through repetition. The fixes are standard: several paraphrases per intent rotated without repeats, two facts aggregated into one sentence, pronouns on second mention, discourse markers ("So", "That gives"), and echoing the user's own wording. N-gram generation is not an upgrade path. A Kneser-Ney 5-gram model reaches perplexity 67.6 on the One Billion Word benchmark against 30.0 for a single LSTM (Chelba et al.; Jozefowicz et al.) **[measured]**, and no n-gram model can keep track of a number from Grove.

Small language models show a sharp quality drop, and it falls where a laptop could otherwise afford them. In Alibaba's own non-thinking-mode table, Qwen3-0.6B scores 6.5 on Arena-Hard (judge-rated open chat) against **36.9 for Qwen3-1.7B**. Multi-turn instruction following (Multi-IF) goes from 33.3 to 44.7, and IFEval from 54.5 to 68.2 (Qwen3 Technical Report, Table 20) **[vendor]**. SmolLM2-360M-Instruct scores 3.66 on MT-Bench against 6.13 for SmolLM2-1.7B-Instruct (SmolLM2 paper; 360M card) **[vendor]**. Google positions Gemma 3 270M for fine-tuned narrow tasks such as classification, extraction and routing, not open chat (VentureBeat). No independent human chat ratings were found for any model below 2B, so all of these are developer claims. TinyStories offers the counterexample that matters. On a deliberately narrow,

simple-vocabulary domain, models under 10M parameters wrote fluent multi-paragraph English. A 28M, 8-layer model scored 9/10 on grammar and consistency in one GPT-4-graded example. One-layer models failed every consistency prompt, and general-web 125M models rarely stayed coherent beyond a few words (Eldan & Li 2023) **[measured]**. Narrowing the domain and cleaning the data buys coherence at tiny scale. Breadth still costs parameters.

The CPU cost of those parameters is well characterised on similar hardware, though not on the 255U. On a Core i7-13700H laptop with llama.cpp at 4-bit, Malakhov measured about **80 tokens/s for Llama-3.2-1B**, 70 for Gemma-3-1B, 40 for the 3.8B Phi-4-mini and 15 for a 7B model. Resident memory was 0.5–0.6 GB for the 1B models and 1.9 GB for Phi-4-mini, and a 256-token prompt took about 3 s to process on the 7B model (Malakhov, CEUR-WS Vol-4164) **[measured]**. Meta reports that 4-bit quantisation gives Llama 3.2 1B about 2.6x faster decoding than bf16 on a phone (model card) **[vendor]**. Hobbyist tests on older Core Ultra chips found the NPU no faster than llama.cpp on the CPU, with model loads of about 96 s (DEV Community) **[community]**. The 255U is a 15 W-class U-series part, below the 45 W i7-13700H, and decoding is mostly bound by memory bandwidth. The planning envelope below is therefore an extrapolation that Stage 0 must replace.

Option	Size / RAM at Q4	Decode on 255U	Quality anchor
Templates and NLG rules	~0	instant	Exact; competitive human-rated quality in E2E **[measured]**
10–50M narrow model, TinyStories-style	<0.1 GB **[estimate]**	hundreds of tok/s **[estimate]**	Fluent in-domain only **[measured, TinyStories]**
135–360M instruct (SmolLM2, Gemma 3 270M)	0.1–0.3 GB **[estimate]**	hundreds of tok/s **[estimate]**	MT-Bench 3.66 at 360M **[vendor]**
0.5–1B (Qwen2.5-0.5B, Qwen3-0.6B, Llama 3.2 1B)	0.4–0.6 GB **[measured on i7-13700H for 1B]**	30–60 tok/s **[estimate]**	Arena-Hard 6.5 at 0.6B **[vendor]**
1.7–2B (Qwen3-1.7B, SmolLM2-1.7B)	1.0–1.3 GB **[estimate]**	20–35 tok/s **[estimate]**	Arena-Hard 36.9, Multi-IF 44.7 **[vendor]**
3–4B (SmolLM3, Qwen3-4B, Phi-4-mini)	~1.9–2.5 GB **[measured for Phi-4-mini on i7-13700H]**	10–20 tok/s; 1–3 s to first token **[estimate]**	IFEval 69–82 **[vendor]**

The useful middle option is narrow. Fine-tune a 135M–360M model only to rephrase a template draft that Grove has already filled. Then re-parse every number in its output and compare it with Grove's result, falling back to the plain template on any mismatch. This applies E2E's semantic-control lesson at a size that costs a fraction of a second. It is an inference, though: no study has measured a rephraser of this kind.

Grove turns a weak program writer into a cautious one

The executor's value is real but specific. In the PAL experiments with Codex, program execution scored 72.0% on GSM8K against 65.6% for chain-of-thought. On GSM-Hard, the same problems with large numbers, the gap widened to **61.2% against 23.1%**. When the model wrote the program but computed the answer itself, accuracy fell to 23.2%, so the gain came from execution. Meaningful variable names were worth about 13 points (59.0 without them against 71.8 with them). Weak code models got worse with programs: text-davinci-001 scored 8.6% with PAL against 26.5% with chain-of-thought (PAL) **[measured]**. For a strong 1.5B maths model, tool use moved GSM8K from 84.8 to 83.7 but MATH from 75.8 to 79.9 (Qwen2.5-Math report) **[vendor]**. An executor removes arithmetic errors. It does not remove setup errors, and setup errors are what SVAMP measures. A small model must therefore be **trained** on the Grove format, as PaD showed, and Grove programs should keep descriptive names.

Selection is the larger lever at small scale, and Grove makes it exact. TinyGSM's 1.3B generator reached **81.5% on GSM8K** with a 1.3B verifier picking among 48 samples, above its GPT-3.5 teacher's 77.4% (TinyGSM) **[measured]**. A secondary summary credits the verifier with about 13 of those points. Choosing the best of eight samples with a reward model added about 8–10 points at 1.5B, more than switching to tool use did

(Qwen2.5-Math) **[vendor]**. Grove does not need a learned verifier for the agreement step. Sampled programs execute to exact fractions, so "three of four samples agree" is an exact comparison, not a fuzzy vote over free text. The system can answer only on agreement and otherwise ask or escalate. The right figure to report is coverage at fixed accuracy, for example "answers 60% of word problems at 99% accuracy", not raw accuracy. The team's Neo results show both sides. On a 40-problem grade-school set, gpt-oss-20b with Grove scored 100% against 97.5% for 120b, at 0.83x the cost. That is consistent with the claim in this report's brief but not statistically significant. On AIME 2025, the same cascade lost (40% against 46.7%), because a program the model writes about its own reasoning verifies only the arithmetic **[measured, team]**. Agreement is only as strong as the independence of the samples.

Grove should also hold the conversation state, because small models cannot. On BFCL function calling, TinyLlama-1.1B produced no calls at all. Qwen3-0.6B scored 72.4 on single-turn calls but about 1.4% multi-turn, and Qwen3-1.7B 8.4%. That paper's tables contradict each other in several cells, so treat the figures as indicative (TinyLLM, arXiv 2511.22138) **[measured]**. SMCalFlow gives the design to copy. Each turn becomes a program that extends a dataflow graph, "refer" and "revise" operations resolve "make it 5 apples instead" against earlier results, and exceptions raised by the engine become the system's next question (Andreas et al. 2020). Applied here, "now double it" is an edit to the current Grove program, and a refusal such as "undefined quantity: cost of a pencil" becomes "How much does one pencil cost?". That is the abstention style the survey literature recommends over a bare "I don't know" (Wen et al., TACL 2025).

Routing completes the hybrid, and Grove gives it an unusually good signal. RouteLLM's routers saved more than 85% of strong-model calls on MT-Bench but only **35% on GSM8K** (LMSYS) **[vendor]**, because maths queries are hard to grade before the cheap model has tried them. FrugalGPT-style cascades avoid that by trying the cheap path first and scoring the result (FrugalGPT). With Grove the scoring function is exact and nearly free. The order is rules, then the embedding classifier, then the structured parser, then the small-model program writer with agreement voting, and only then a clarifying question or escalation. The router no longer needs to predict difficulty. It needs to send small talk away from the engine, because a front end that over-produces maths calls will feed greetings to Grove.

A few thousand validated replies and a test suite beat big corpora and BLEU

The reply bank is a curation problem before it is a data problem, and licences narrow the choice sharply. The best-fitting human small-talk corpora are non-commercial: DailyDialog is CC BY-NC-SA 4.0 and EmpatheticDialogues is CC BY-NC 4.0 (DailyDialog card; EmpatheticDialogues LICENSE). PersonaChat, Blended Skill Talk, Cornell, Ubuntu, OpenSubtitles and Reddit dumps carry no clear licence, and DailyDialog and OpenSubtitles also leak test dialogues into training splits (Wen, Luo & Mou 2022). The commercially safe pool is synthetic or task-oriented. SODA is CC BY 4.0, million-scale, and rated more natural than DailyDialog by its authors (allenai/soda). Synthetic-Persona-Chat is CC BY 4.0 (card). OASST1 is Apache 2.0, with over 10,000 human-written conversation trees (card). For the tutoring register, MathDial is CC BY 4.0, with 2,861 teacher–student conversations on multi-step word problems and every teacher move labelled. Its authors note that LLM tutors tend to leak solutions early, while human teachers scaffold with questions (MathDial; Macina et al. 2023). UltraChat is MIT-licensed but describes itself as research-only. Treat these sets as sources of phrasings to hand-edit, not as banks to ship raw.

Smart Reply's build process scales down to a two-person team. Google canonicalised frequent replies by dropping modifiers, so near-identical "thanks" variants collapsed into one. It seeded about 100 intent clusters with 3–5 labelled examples each, expanded them by label propagation, and had raters validate cluster members. At runtime it showed at most one reply per intent. Removing that diversity rule cut click-through by about 7.5% relative, and daily use spanned 12.9K unique suggestions in 376 clusters (Kannan et al. 2016) **[vendor]**. XiaoIce kept its persona consistent by filtering the candidate pool for persona fit, not by ranking alone. It replaced "I don't know" with editorial lines such as "Hmmm, difficult to say. What do you think?"

(Zhou et al. 2020). For this system that means a single persona sheet, fixed hand-written answers to persona and backstory questions, and a bank of a few dozen intents with 5–10 validated variants each, plus ChatScript-style gambits and rejoinders around maths topics. That is a few hundred to about 3,000 entries **[estimate]**. No source measured how many variants users need before they notice repetition.

Evaluation should be behavioural and paired, never BLEU. In USR's analysis, BLEU-4 correlated with human turn-level quality judgements at only 0.14 (Spearman) on PersonaChat, against 0.48 for USR itself (Mehri & Eskenazi 2020). A study of 23 learned metrics found most work best on the datasets they were built for (Yeh et al. 2021) **[measured]**. CheckList-style suites fit a template bot well. Minimum-functionality tests check that "hi" and "I don't get it" route correctly. Invariance tests check that typos, casing and politeness padding change neither the intent nor the maths answer. Directional tests check that "explain slowly" adds steps. Practitioners using CheckList wrote twice as many tests and found almost three times as many bugs (Ribeiro et al. 2020) **[measured]**. Retrieval gets Recall@1 and Recall@3 on held-out real messages. Conversational quality gets ACUTE-Eval: blinded pairwise comparisons of whole conversations, one question per axis, tested with a binomial test. Likert ratings anchor and drift between runs (Li, Weston & Roller 2019). With 40 pairs per axis, a version must win **27 of 40 (68%)** to clear a two-sided 5% binomial test (computed), so small panels detect only clear improvements. An LLM judge, for example Neo's gpt-oss over the API, is acceptable for triage if each pair is run in both orders, since even GPT-4 kept its verdict after a swap only 65% of the time. It must never grade maths: GPT-4 accepted wrong maths answers 14 times in 20 without a reference solution (Zheng et al. 2023) **[measured]**. Grove or gold answers grade every number. XiaoIce warns that turns-per-session can be gamed by bland, prodding replies. A failure log of wrong intent, repeated reply, fallback fired, persona contradiction and maths error, with every entry turned into a regression test, is the engagement metric a small team can trust.

A six-stage plan, from measuring the laptop to an optional 1.7B fallback

The stages are ordered so that each one ships alone and the cheapest component handles the most traffic. The first three need no model larger than about 30M parameters. Stage 3 adds a 60M model, Stage 4 a sub-1B model on fallback traffic only, and Stage 5 is optional. Effort is in person-weeks for a two-person team, and every effort figure is an **[estimate]**.

Stage	Added CPU latency per turn	Memory	Data	Effort
0. Measure and freeze tests	none	none	300–500 labelled messages, SVAMP, scripted dialogues	1 week
1. Dialogue manager, router, hand-written bank	<5 ms router and rules; autocorrect ~3 ms/word **[measured, team]**, reducible by OOV gating	<100 MB **[estimate]**	10–30 examples per intent; 300–1,000 replies	3–4 weeks
2. Retrieval over a curated bank	1–20 ms **[estimate]**	<300 MB **[estimate]**	2–10K replies from licence-clean sources	2–3 weeks
3. Word problems: parser + 60M program writer	50–500 ms with k samples **[estimate]**	~0.3 GB fp32 **[estimate]**	Verified Grove programs from GSM8K, MAWPS, ASDiv	4–6 weeks
4. Sub-1B program writer as fallback	2–10 s on the fallback slice only **[estimate]**	~0.4 GB at Q4 **[estimate]**	Same, plus self-training data	2–3 weeks
5. Rephraser and open-question fallback (optional)	0.2–1 s rephraser; 3–10 s with a 1.7B model **[estimate]**	0.2–1.3 GB **[estimate]**	Draft→reply pairs; persona sheet	3–5 weeks

Stage 0 replaces estimates with measurements on the 255U

Run `llama-bench` CPU-only with physical-core thread counts, reporting pp512 and tg128 separately (Malakhov used the same harness), for Qwen2.5-0.5B, Qwen3-1.7B and a 135M–360M model. Time batch-size-1 encoding for potion-base-8M and MiniLM-L6, profile the autocorrect and Grove per turn, and record p50 and p95 for each. Freeze the test assets: 300–500 real or scripted user messages labelled with acceptable intents, including out-of-scope items; a CheckList suite; SVAMP and a GSM8K sample; and 20 scripted multi-turn conversations for ACUTE pairs. The current system's scores on all of them become the baseline. This stage **achieves** a measured cost table and a frozen yardstick. It **does not** change behaviour. It is **measured** by whether every later stage reports against these sets. At SVAMP's n = 1,000, one standard error is about 1.2 points at 18% accuracy and 1.5 points at 40% (computed), so differences under about 4 points are noise.

Stage 1 builds the ChatScript-style core

Add a two-layer router. Rules and regular expressions catch digits, maths cue words, greetings and commands first. A potion-base-8M plus logistic-regression classifier sorts the rest into small talk, maths, help, persona and out-of-scope, with a probability threshold for abstention. Behind it sit topics with gambits and rejoinders, a persona sheet with fixed backstory answers, ChatScript's 20-volley no-repeat memory, editorial fallbacks that offer an example problem instead of "I don't know", and template NLG with rotated paraphrases, aggregation, pronouns and discourse markers. Turn Grove's refusals into targeted clarifying questions. **Achieves:** bot-led, persona-consistent small talk around maths, help and onboarding, with exact arithmetic in friendlier English. **Does not achieve:** word-problem understanding, general knowledge or open chat. Users who leave the scripted topics hit fallbacks. **Measure:** intent accuracy and out-of-scope recall on the held-out set (target at least 90% on in-scope intents, plausible from the narrow-domain NLU results above **[estimate]**), CheckList pass rate, repeated replies per 20 turns, and ACUTE pairs against the current templates.

Stage 2 widens coverage with retrieval

Harvest candidate replies from SODA, Synthetic-Persona-Chat, OASST1 and MathDial tutor moves. Canonicalise, cluster by intent with the embedding model, validate by hand against the persona sheet, and index with BM25 or MiniLM. At runtime, retrieve, keep at most one reply per intent, skip anything recently used, and abstain to Stage 1 fallbacks below a similarity threshold. **Achieves:** broader small-talk coverage with human-written fluency, Smart Reply style. **Does not achieve:** real multi-turn context, facts, or anything beyond the bank. Coverage gaps remain, and XiaoIce needed a generator to patch them. **Measure:** Recall@1 and Recall@3 on held-out messages, abstention rate, failure-log rate per 100 turns, and ACUTE pairs against Stage 1.

Stage 3 teaches the system to read word problems without an LLM

Build two program writers and keep whichever gives more coverage at 98–99% precision. The structured arm uses the spaCy small parse, a number normaliser, Hosseini-style verb categories and Roy-and-Roth relevance and operation classifiers. It explicitly parses the question sentence for the target entity and unknown, then runs a beam search over Grove programs under positivity and integrality constraints. The learned arm replaces cpulm's from-scratch model with a pretrained CodeT5-small (60M), fine-tuned PaD-style on execution-verified Grove programs. The grovelm data builder already converts GSM8K, MAWPS and ASDiv into verified Grove programs. Both arms sample or beam several programs, execute them in Grove, answer on agreement, and otherwise ask a clarifying question. Multi-turn edits ("what if there were 5?") become revisions of the current program. **Achieves:** genuine word-problem handling on simple arithmetic. A realistic SVAMP range is 35–50%, anchored on Graph2Tree's 43.8% and PaD's 32.2% GSM8K at 60M **[estimate]**, delivered at high precision because disagreement triggers abstention. **Does not achieve:** robustness on problems with three or more quantities or inverted operations, or anything beyond Grove's expressivity (units, dates, geometry). **Measure:** SVAMP and GSM8K accuracy and coverage-at-precision curves, split by number of quantities and SVAMP variation type, plus the share of abstentions that the clarifying question resolves.

Stage 4 adds a sub-1B program writer only where Stage 3 abstains

Run the grovelm prototype on Colab as planned (Qwen2.5-0.5B-Instruct with LoRA, runs R0 base, R1 SFT, R2 one self-training round), quantise to Q4, and call it only on problems Stage 3 abstains on. Sample up to eight programs with early exit once two or three agree, and show a "working on it" message while it runs. If 0.5B falls short, Qwen2.5-1.5B (73.2% GSM8K **[vendor]**, about 1 GB) is the next step. Verified outputs feed checker-based self-training, the "checker trains the model" loop. **Achieves:** the largest single jump in maths understanding available on CPU. The anchors are PaD's 51.0% SVAMP at 0.77B and TinyGSM's 81.5% GSM8K at 1.3B plus a verifier **[measured]**. It strains the "no LLM" ideal only weakly, since the model is 0.5B and runs locally on fallback traffic. **Does not achieve:** chat, since the model only writes programs, or fast answers on its slice, since 2–10 s per hard problem **[estimate]** must be confirmed in Stage 0. **Measure:** accuracy and coverage of the cascade against Stage 3 alone on frozen SVAMP and GSM8K, fallback rate, and p95 latency on the fallback slice.

Stage 5 is optional: natural phrasing and an honest boundary for general questions

There are two separable additions. First, a 135M–360M model fine-tuned to rephrase Grove-filled template drafts. Training pairs can come from the team's own templates rewritten by gpt-oss through Neo, after checking the teacher model's terms. A numeric-fidelity check rejects any rephrasing whose numbers differ from Grove's. Second, for general questions, choose deliberately among three options: deflect within the persona; run Qwen3-1.7B locally, at about 1.0–1.3 GB and 20–35 tok/s **[estimate]**, with Grove on every number and a visible hedge; or escalate to Neo when the user is online. **Achieves:** less template-like phrasing, and some answer to "what is photosynthesis?". **Does not achieve:** reliable facts. No source here shows a sub-2B model answering general questions without hallucinating, and this is the one place where the system becomes LLM-dependent. **Measure:** ACUTE pairs against Stage 2 on naturalness, the fidelity check's rejection rate (zero mismatched numbers may reach users), and a hand audit of 100 general-question answers for factual errors.

Conclusion

The research separates two problems that look like one. **Conversation is an authoring problem** with a known shape: bot-led topics, a validated bank, a no-repeat memory and good fallbacks. Its cost is human weeks, not CPU or parameters, and the pre-LLM record says it tops out around 3.5–4 out of 5 on open small talk. **Comprehension is a modelling problem**, where cpulm's 18.0% shows that a from-scratch model at that size learns only surface cues. The next gains come from pretraining plus program-format training (60M–0.5B), not from more rules or a bigger from-scratch model. Budget each accordingly: write content for chat, and fine-tune small pretrained models for maths.

Grove's refusal is the asset that makes small models acceptable. A cheap model that knows when to stay silent and ask a question is worth more than a larger one that answers everything, so the target metric should shift from SVAMP accuracy to **coverage at 99% precision**, which only an exact executor can report honestly. The one real boundary is general knowledge. Below about 1.7B parameters no evidence supports open-domain answers, so the decision is about product scope (a maths companion with a personality, or a general assistant) and not a technical gap that more engineering will close cheaply.